

Kiwi Syslog Server vs SyslogAgent Desktop

Gap Analysis Report — Missing Features & Functionalities | March 2026

Purpose: This document lists the features available in Kiwi Syslog Server that are currently **not implemented** in SyslogAgent Desktop. These are grouped by priority to help plan the development roadmap.

1. High Priority — Core Features

These are fundamental Kiwi capabilities that most users expect. Missing these would be noticed immediately when switching from Kiwi.

Feature	What Kiwi Does	Why It Matters
TLS / SSL Encryption	Sends and receives syslog over TLS (RFC 5425). Supports certificate-based authentication and encrypted transport.	Without TLS, all log data travels in cleartext. Most compliance frameworks (PCI-DSS, HIPAA, ISO 27001) require encrypted log transport.
RFC 5424 Syslog Format	Supports the modern structured syslog format with structured data elements, message IDs, and version headers.	SyslogAgent only supports RFC 3164 (legacy). Many modern SIEMs (Splunk, Elastic, QRadar) prefer or require RFC 5424.
Multiple Destination Servers	Forwards logs to 2 or more syslog servers at the same time. Supports primary/failover configuration.	Currently SyslogAgent supports only 1 target server. If that server goes down, all logs are lost with no failover.
Syslog Receiver Mode	Acts as a syslog server — receives logs from network devices (routers, switches, firewalls, printers, IoT devices).	SyslogAgent is send-only. It cannot collect logs from network equipment the way Kiwi does as a central syslog server.
Local File Logging	Writes collected/received logs to local files on disk with daily or size-based rotation.	SyslogAgent has no local log persistence. If the remote syslog server is unreachable, events are permanently lost.

Message Queue / Retry Buffer	Queues messages in memory or on disk when the destination is offline. Automatically replays them when connectivity is restored.	Currently, if a send fails in SyslogAgent, the event is dropped. There is no retry or buffering mechanism.
Windows Service Mode	Runs as a native Windows Service (SCM-managed), starts automatically on boot, survives user logoff.	SyslogAgent runs as a desktop app. It stops when the user logs off, making it unsuitable for 24/7 server monitoring.

2. Medium Priority — Enterprise Features

These features are commonly used in enterprise Kiwi deployments for alerting, integration, and operational management.

Feature	What Kiwi Does	Why It Matters
Email Alerts	Sends email notifications when specific events match defined rules (e.g., 5+ failed logons in a row, critical errors).	Allows real-time alerting to admins without needing a separate SIEM or monitoring tool.
SNMP Trap Reception	Receives SNMP traps from network devices and converts them into syslog messages for unified logging.	Network devices (switches, routers, UPS) often only support SNMP. Without this, those devices can't be monitored.
SNMP Trap Forwarding	Forwards matched events as SNMP traps to monitoring tools like Nagios, PRTG, or Zabbix.	Enables integration with existing network monitoring infrastructure.
Database Output	Writes logs directly to SQL Server, MySQL, PostgreSQL, or any ODBC-compatible database.	Enables long-term log storage, SQL-based querying, and integration with reporting tools.
CEF / LEEF Format Support	Outputs logs in Common Event Format (for ArcSight) and LEEF (for IBM QRadar).	Required for direct integration with enterprise SIEM platforms that expect these formats.
Log Compression & Archival	Compresses rotated log files (gzip) and archives them with retention policies (e.g., keep 90 days).	Reduces disk usage for long-term storage and supports compliance retention requirements.

Rules Engine / Actions	Complex rule chains: "If event matches condition X, then do action Y" — forward, alert, drop, rewrite, run script.	Kiwi's core value proposition. Allows flexible event routing and response without external tools.
Persistent Statistics	Stores historical statistics — messages/sec, per-source counts, error rates — with graphs over time.	SyslogAgent metrics reset on restart. No way to track trends or prove SLA compliance.
Scheduled Reports	Generates and emails daily/weekly reports of log volumes, top sources, top event types.	Provides management visibility without requiring manual log review.

3. Lower Priority — Nice-to-Have Features

These are useful but not essential for a Kiwi replacement. They can be deferred to later phases.

Feature	What Kiwi Does	Why It Matters
DNS Resolution	Resolves source IP addresses to hostnames in received log messages for easier reading.	Convenience feature — makes logs more human-readable.
Message Rewriting	Regex-based find/replace to transform or enrich messages before forwarding.	Useful for normalizing logs from different sources into a consistent format.
IP-based Filtering	Accept or reject messages based on source IP address or subnet.	Only relevant when acting as a syslog receiver (depends on receiver mode being implemented first).
Custom Scripting	Runs VBScript, JScript, or PowerShell scripts as actions when events match rules.	Enables custom automation — e.g., disable an account, block an IP, create a ticket.
Web Management Console	Browser-based admin UI for configuration, monitoring, and rule management.	Enables remote management without RDP. SyslogAgent currently requires desktop access.
JSON Structured Output	Forwards events as structured JSON for modern log pipelines (Elastic, Splunk HEC, Loki).	Modern observability stacks prefer JSON over traditional syslog formats.
High Availability	Primary/standby agent failover, clustered deployment for zero-downtime log collection.	Enterprise requirement for mission-critical environments.

4. Recommended Implementation Order

Based on impact and effort, here is the suggested order to close the gap with Kiwi:

Order	Feature	Priority	Reason
1	Windows Service Mode	High	Without this, the agent stops when the user logs off. This is a basic requirement for any production log agent.
2	TLS / SSL Encryption	High	Required by most security compliance standards. Straightforward to add using .NET

			SslStream.
3	Message Queue / Retry	High	Prevents silent event loss during network outages. Can start with a simple in-memory queue with disk spillover.
4	Multiple Destinations	High	At minimum, support a primary + failover server for redundancy.
5	Local File Logging	High	Fallback storage when the remote server is unavailable. Also useful for local audit/debug.
6	RFC 5424 Format	High	Modern SIEMs expect it. Can be added alongside existing RFC 3164 as a config toggle.
7	Rules Engine	Medium	Core Kiwi differentiator. Enables flexible event routing and response.
8	Email Alerts	Medium	Quick win for operational visibility once the rules engine is in place.
9	Syslog Receiver Mode	Medium	Needed to fully replace Kiwi as a central syslog server for network devices.
10	Database Output	Medium	Enables long-term queryable storage and reporting.

5. Summary

- **7 High-Priority** features are missing — these are needed for production readiness and Kiwi parity.
- **9 Medium-Priority** features are missing — these cover enterprise use cases (alerting, SIEM integration, reporting).
- **7 Lower-Priority** features are missing — these are nice-to-have and can be deferred.
- **Total: 23 features** to reach full Kiwi Syslog Server parity.

SyslogAgent Desktop already covers the core log collection well (Windows Event Logs, Linux Journald, file watchers, filtering, deduplication, real-time + batch modes). The main gaps are around **transport security, resilience, and enterprise integrations**.